Hysteresis and Reliability of hBN-encapsulated MoS_2 FETs

F. de los Santos-Prieto*, A. Verdianu*, A. Provias *, D. Polyushkin†, D. Waldhoer*, T. Müller†, T. Grasser*, and T. Knobloch*,

* Institute for Microelectronics, Technical University of Vienna, Vienna, Austria

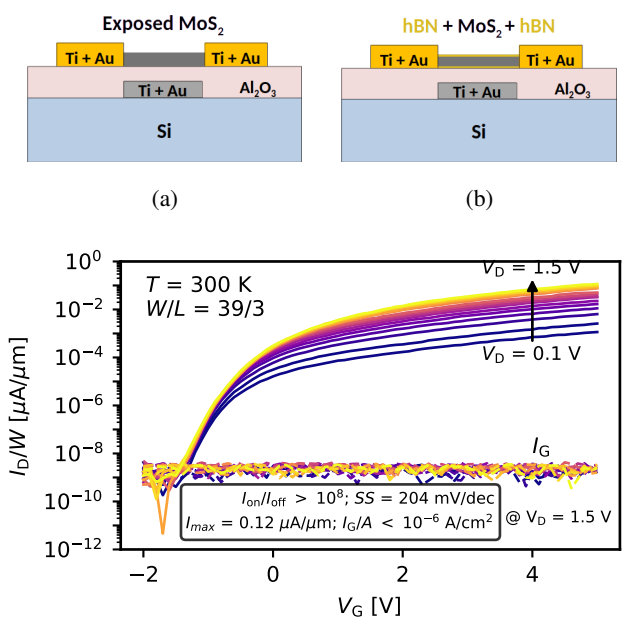
† Institute of Photonics, Technical University of Vienna, Vienna, Austria

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

II. Methods and Fabrication



(a) (b)

(c)

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and main device metrics.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

We gratefully acknowledge funding from the European Innovation Council under project number 101223249.

References

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [4]. Papers that have been accepted for publication should be cited as "in press" [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

References

[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, "Title of paper if known," unpublished.

[5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove the template text from your paper may result in your paper not being published.

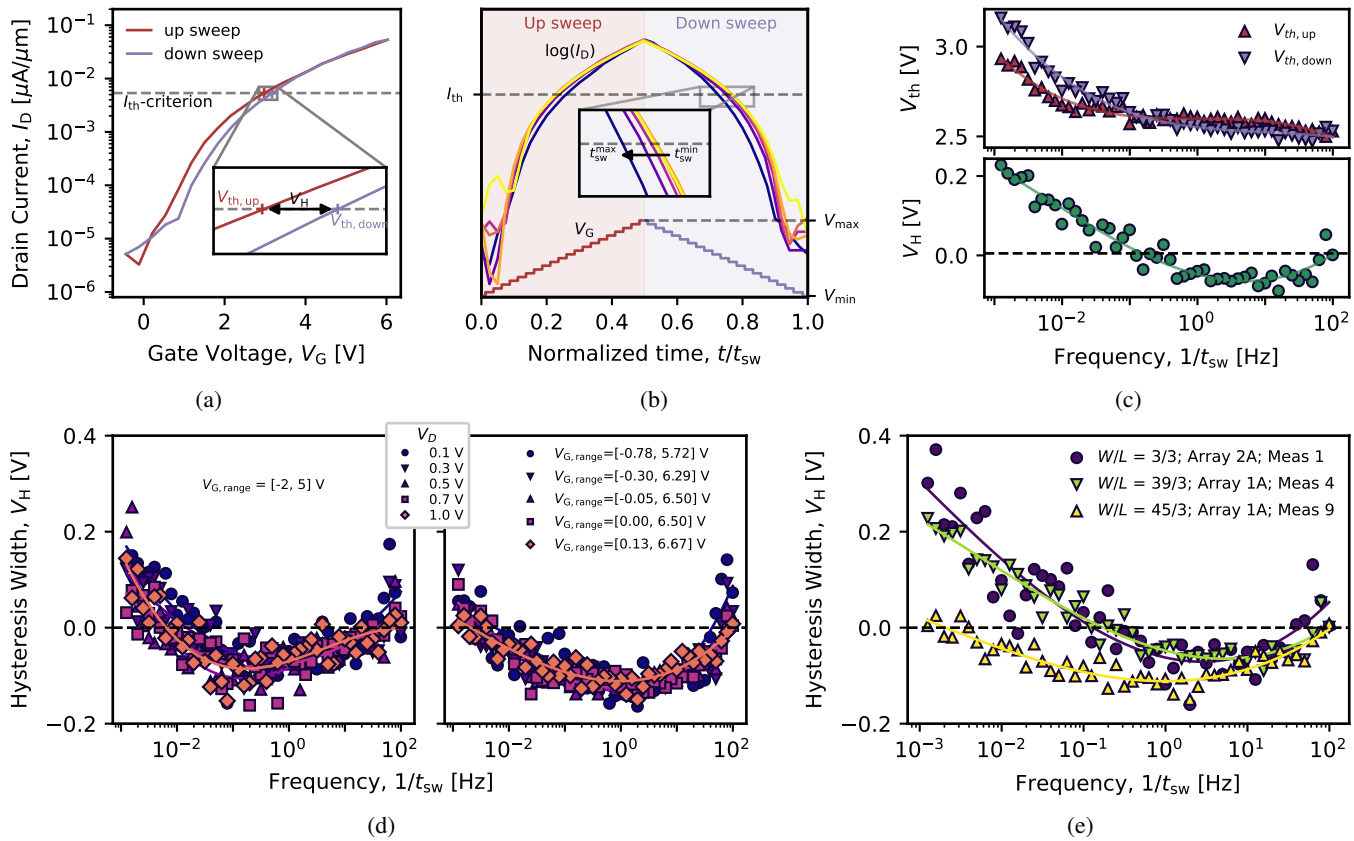
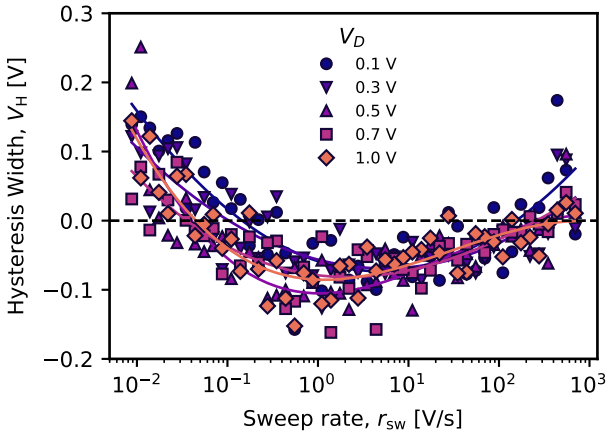
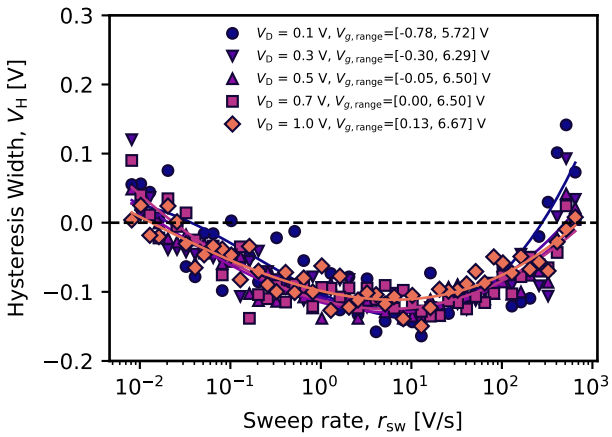


Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_{H}, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{\mathrm{th,up}} - V_{\mathrm{th,down}}$. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_{G} are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep rates, r_{sw}. (c) **Hysteresis-r_{sw} curves**: Threshold voltages, $V_{\mathrm{th,up}}$ and $V_{\mathrm{th,down}}$, and hysteresis width, V_{H}, are shown as a function of the sweep rate, r_{sw}.

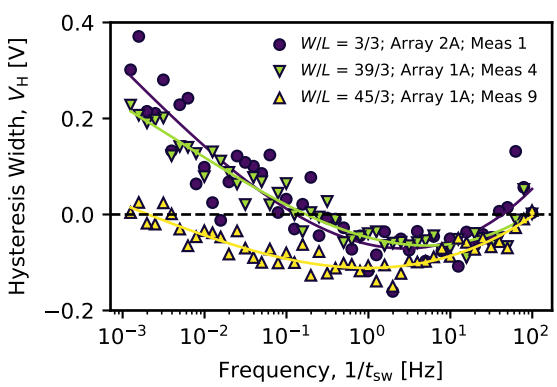


(a)



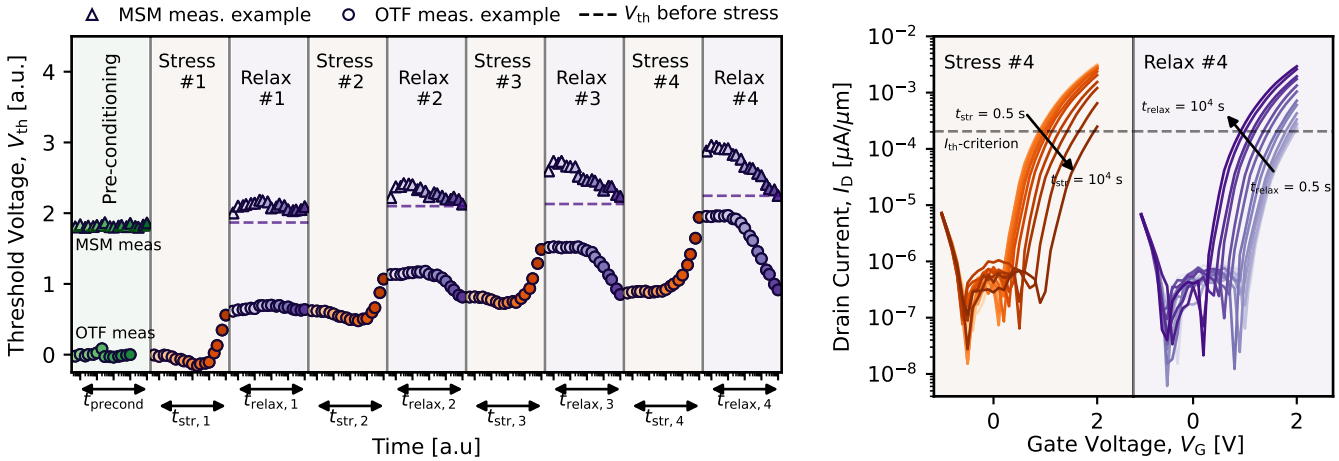
(b)

Fig. 3. (a) Fixed gate voltage range: hysteresis measurements for different drain voltages with a fixed gate voltage range, $V_{G,range}$ = [-2,5] V. (b) Adjusted gate voltage range for constant overdrive electric field: hysteresis measurements for different drain voltages with $V_{G,range}$ adjusted to maintain a constant overdrive electric field, E_{od} = 3 MV/cm.



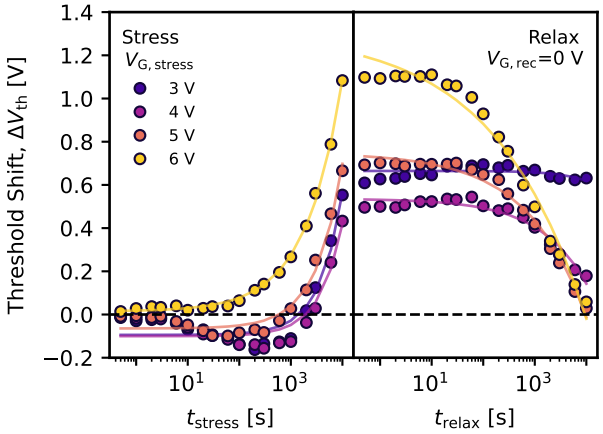
(a)

Fig. 4. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.



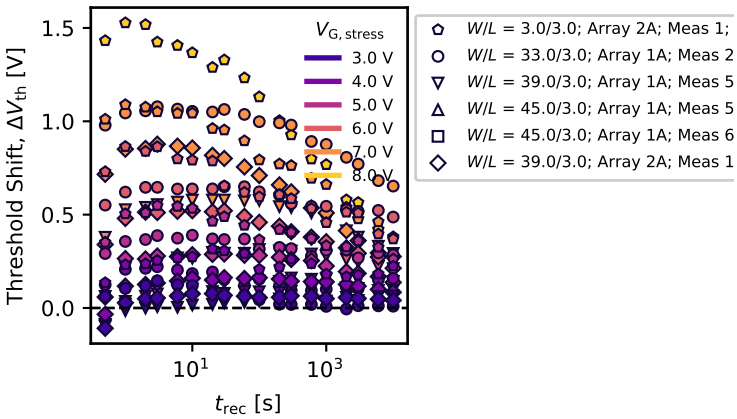
(a) (b)

Fig. 5. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.

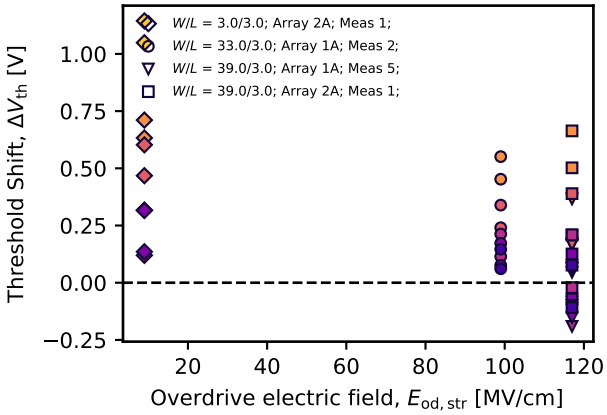


(a)

Fig. 6. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.



(a)



(b)

Fig. 7. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.